J. R. Statist. Soc. B (2003)
65, *Part* 2, *pp.* 367–389

Distributions generated by perturbation of symmetry with emphasis on a multivariate skew *t*-distribution

Adelchi Azzalini

Università di Padova, Italy

and Antonella Capitanio

Università di Bologna, Italy

[Received April 2001. Final revision October 2002]

Summary. A fairly general procedure is studied to perturb a multivariate density satisfying a weak form of multivariate symmetry, and to generate a whole set of non-symmetric densities. The approach is sufficiently general to encompass some recent proposals in the literature, variously related to the skew normal distribution. The special case of skew elliptical densities is examined in detail, establishing connections with existing similar work. The final part of the paper specializes further to a form of multivariate skew t-density. Likelihood inference for this distribution is examined, and it is illustrated with numerical examples.

Keywords: Asymmetry; Central symmetry; Elliptical distributions; Healy's plot; Multivariate t-distribution; Quadratic forms; Skewness; Skew normal distribution

1. Introduction

1.1. Motivation and aims

There is growing interest in the literature on parametric families of multivariate distributions which represent, in some sense, departures from the multivariate normal family. The motivation of these efforts is to introduce more flexible parametric families that are capable of adapting as closely as possible to real data, in particular in the quite frequent case of phenomena whose empirical outcome behaves in a non-normal fashion but still retains some broad similarity with the multivariate normal distribution. Typically this departure from normality occurs in the form of a roughly bell-shaped density, but with contour levels that are not quite elliptically shaped and/or with contour levels that are not quite spaced as the normal density prescribes.

Some of this literature is connected with the so-called multivariate skew normal (SN) distribution, which has recently been studied by Azzalini and Dalla Valle (1996) and Azzalini and Capitanio (1999); this has been further developed by other researchers whose work will be referenced later in this section. The d-dimensional SN density, in the 'standard' form which does not include location and scale parameters, is

$$2\,\phi_d(y;\bar{\Omega})\,\Phi(\alpha^{\mathrm{T}}y), \qquad y \in \mathbb{R}^d, \tag{1}$$

Address for correspondence: Adelchi Azzalini, Dipartimento di Scienze Statistiche, Università degli Studi di Padova, via Cesare Battisti 241, 35121 Padova, Italy.
E-mail: azzalini@stat.unipd.it

 1369–7412/03/65367

where $\phi_d(y; \bar{\Omega})$ is the $N_d(0, \bar{\Omega})$ density at y for some correlation matrix $\bar{\Omega}$, $\Phi(\cdot)$ is the $N(0,1)$ distribution function and $\alpha \in \mathbb{R}^d$. Here α plays the role of a shape parameter; when $\alpha = 0$, we recover the regular normal density.

As a further level of generalization of the normal distribution, Azzalini and Capitanio (1999), page 599, presented a lemma which leads to the construction of a 'skew elliptical' density, which is an elliptical density multiplied by a suitable skewing factor, in such a way that the product is still a proper density. Branco and Dey (2001) considered another form of skew elliptical distribution, whose connections with the one mentioned above will be discussed extensively in this paper. Other work on extensions of elliptical families has been done by Genton and Loperfido (2002), where it is shown that distributional properties of certain functions of elliptical variates extend to their skewed variants, generalizing a similar result of Branco and Dey (2001).

Arnold and Beaver (2000a) studied a variant of expression (1) which replaces the argument of Φ by $\alpha_0 + \alpha^{\mathrm{T}} y$, where α_0 is an additional parameter, with consequent adjustment of the normalizing constant. The same variant of the SN distribution has been considered by Capitanio *et al.* (2003) in the context of graphical models. Sahu *et al.* (2001) studied yet another form of skew elliptical distribution, where the skewing factor is a d-dimensional distribution function, rather than a scalar function like those of the previously mentioned cases. In the same spirit as expression (1), Arnold and Beaver (2000b) studied a form of multivariate skew Cauchy distribution. For additional references and a recent review on the connected literature, see Arnold and Beaver (2002).

One purpose of the present contribution is to propose a fairly general extension of expression (1); in addition, a better understanding of the connections and similarities between some of the above-described proposals is attempted. A broad formulation is presented in Section 2 and is specialized to a skew elliptical form in Section 3. This approach encompasses several of the existing proposals and it appears to provide a potentially general framework for special cases. We discuss in detail a few of these and, from Section 4 onwards, we focus on a form of multivariate skew t-distribution. Since this represents mathematically quite a manageable distribution, allowing ample flexibility in skewness and kurtosis, it appears to be a promising tool for a wide range of practical problems. Associated likelihood inference for this skew t-distribution and illustrative examples are presented in Section 5. The final section provides a brief general discussion. Some background information on the SN distribution and the elliptical family is given in the second part of this introductory section.

For brevity various results and proofs that were included in an earlier version of the paper had to be removed. The full version of the paper is available at `http://www.stat.unipd.it/~azzalini/SN`, where one can also obtain the freely available software referred to in the text.

1.2. Some preliminaries

1.2.1. The skew normal distribution

Given a full rank $d \times d$ covariance matrix $\Omega = (\omega_{rs})$, define

$$\omega = \mathrm{diag}(\omega_1, \ldots, \omega_d) = \mathrm{diag}(\omega_{11}, \ldots, \omega_{dd})^{1/2}$$

and let $\bar{\Omega} = \omega^{-1}\Omega\omega^{-1}$ be the associated correlation matrix; also let $\xi, \alpha \in \mathbb{R}^d$. A d-dimensional random variable Z is said to have an SN distribution if it is continuous with density function at $z \in \mathbb{R}^d$ of type

$$2\,\phi_d(z - \xi; \Omega)\,\Phi\{\alpha^{\mathrm{T}}\omega^{-1}(z - \xi)\}. \tag{2}$$

We shall then write $Z \sim \mathrm{SN}_d(\xi, \Omega, \alpha)$, referring to ξ, Ω and α as the location, dispersion and shape or skewness parameters respectively. Density (1) corresponds to the 'standard' distribution $\mathrm{SN}_d(0, \bar{\Omega}, \alpha)$.

By varying α, we obtain a variety of shapes; Azzalini and Dalla Valle (1996) displayed graphically some instances of them when $d = 2$. Clearly, when $\alpha = 0$, we are back to the $N_d(\xi, \Omega)$ density. The cumulant-generating function is

$$K_Z(t) = t^{\mathrm{T}}\xi + \tfrac{1}{2}t^{\mathrm{T}}\Omega t + \zeta_0(\delta^{\mathrm{T}}\omega t)$$

where

$$\begin{aligned} \delta &= \frac{1}{(1+\alpha^{\mathrm{T}}\bar{\Omega}\alpha)^{1/2}}\bar{\Omega}\alpha, \\ \zeta_0(x) &= \log\{2\,\Phi(x)\}. \end{aligned} \tag{3}$$

From the expression for δ we have

$$\alpha = \frac{1}{(1-\delta^{\mathrm{T}}\bar{\Omega}^{-1}\delta)^{1/2}}\bar{\Omega}^{-1}\delta. \tag{4}$$

There are at least two stochastic representations for Z. These are useful for random-number generation and for deriving in a simple way several formal properties.

(a) *Conditioning method*: suppose that U_0 is a scalar random variable and U is a d-dimensional variable, such that

$$\begin{aligned} \begin{pmatrix} U_0 \\ U \end{pmatrix} &\sim N_{d+1}(0, \Omega^*), \\ \Omega^* &= \begin{pmatrix} 1 & \delta^{\mathrm{T}} \\ \delta & \bar{\Omega} \end{pmatrix} \end{aligned} \tag{5}$$

where Ω^* is a full rank correlation matrix. Then the distribution of $(U|U_0 > 0)$ is $\mathrm{SN}_d(0, \bar{\Omega}, \alpha)$ where α is a function of δ and $\bar{\Omega}$; in fact, we can also set

$$Z = \begin{cases} U, & \text{if } U_0 > 0, \\ -U, & \text{if } U_0 < 0. \end{cases}$$

By an affine transformation of the resulting variable we obtain a distribution of type (2).

(b) *Transformation method*: suppose now that

$$\begin{pmatrix} U_0' \\ U' \end{pmatrix} \sim N_{d+1}\left\{0, \begin{pmatrix} 1 & 0 \\ 0 & \Psi \end{pmatrix}\right\} \tag{6}$$

where Ψ is a full rank correlation matrix, and define

$$Z_j = \delta_j|U_0'| + (1-\delta_j^2)^{1/2}U_j', \tag{7}$$

where $-1 < \delta_j < 1$ for $j = 1, \ldots, d$. Then $(Z_1, \ldots, Z_d)$ has the d-dimensional SN distribution, with parameters which are suitable functions of the δs and Ψ.

A third type of representation is known to exist in the scalar case. If (U_0, U_1) is a bivariate normal variate with standardized marginals and correlation ρ, then

$$\max(U_0, U_1) \sim \mathrm{SN}(0, 1, \alpha) \tag{8}$$

where $\alpha = \{(1-\rho)/(1+\rho)\}^{1/2}$. This result has been given by Roberts (1966), in an early explicit occurrence of the scalar SN distribution, and later rediscovered by Loperfido (2002); the same conclusion can also be obtained from a result of H. N. Nagaraja, quoted by David (1981), exercise 5.6.4. The generalization of this type of representation to the multivariate setting to obtain density (1) via a set of $\max(\cdot)$ operations on normal variates is an open question.

Among the many formal properties that are shared with the normal class, a noteworthy fact is that

$$(Z-\xi)^{\mathrm{T}}\Omega^{-1}(Z-\xi) \sim \chi^2_d. \tag{9}$$

Other properties of quadratic forms of SN variables are given by Azzalini and Capitanio (1999), Genton *et al.* (2001) and Loperfido (2001). Another important property of this class is closure under affine transformations of the variable Z; in particular, this implies closure under marginalization, i.e. the distribution of all subvectors of Z is still of type (2).

What is lacking is closure under conditioning, i.e. the conditional distribution of a set of components of Z given another set of components is not of type (2). This property is achieved by a simple extension of expression (2) which has been examined by Arnold and Beaver (2000a) and by Capitanio *et al.* (2003). This variant of the density takes the form

$$\Phi(\tau)^{-1}\,\phi_d(z-\xi;\Omega)\,\Phi\{\alpha_0+\alpha^{\mathrm{T}}\omega^{-1}(z-\xi)\} \tag{10}$$

where τ ($\tau \in \mathbb{R}$) is an additional parameter and

$$\alpha_0 = (1-\delta^{\mathrm{T}}\bar{\Omega}^{-1}\delta)^{-1/2}\tau.$$

When $\tau=0$, $\alpha_0=0$ and expression (10) reduces to density (2). Unfortunately, the χ^2-property (9) does not hold for density (10), if $\tau \neq 0$. A form of genesis of density (10) via conditioning using expression (6) is by consideration of $(U|U_0+\tau>0)$.

1.2.2. Elliptical distributions

We summarize briefly a few concepts about and establish notation for elliptical distributions, confining ourselves to random variables without discrete components. For a full treatment of this topic, we refer the reader to Fang *et al.* (1990).

A d-dimensional continuous random variable Y is said to have an elliptical density if this is of the form

$$f(y;\xi,\Omega) = \frac{c_d}{|\Omega|^{1/2}}\tilde{f}\{(y-\xi)^{\mathrm{T}}\Omega^{-1}(y-\xi)\}, \qquad y \in \mathbb{R}^d,$$

where $\xi \in \mathbb{R}^d$, Ω is a covariance matrix, $\tilde{f}$ is a suitable function from $\mathbb{R}^+$ to $\mathbb{R}^+$, called the 'density generator', and c_d is a normalizing constant. We shall then write $Y \sim \mathrm{Ell}_d(\xi,\Omega,\tilde{f})$.

The basic case is obtained by setting $\tilde{f}(x)=\exp(-x/2)$ and $c_d=(2\pi)^{-d/2}$, leading to the multivariate normal density. Two other important special cases, which will be used extensively in what follows, are provided by the multivariate Pearson type VII distributions, whose generator and normalizing constant are

$$\tilde{f}(x) = (1+x/\nu)^{-M},$$
$$c_d = \frac{\Gamma(M)}{(\pi\nu)^{d/2}\,\Gamma(M-d/2)},$$

where $\nu>0$ and $M>d/2$, and by the multivariate Pearson type II distributions for which

$$\tilde{f}(x) = (1-x)^{\nu},$$
$$c_d = \frac{\Gamma(d/2+\nu+1)}{\pi^{d/2}\,\Gamma(\nu+1)}$$

where $0 \leqslant x \leqslant 1$ and $\nu > -1$. The special importance of the type VII distribution lies in that it includes the multivariate t-density when $M = (d+\nu)/2$, and hence also the Cauchy distribution. For these distributions, we shall use the notation $\mathrm{PVII}_d(\xi, \Omega, M, \nu)$ and $\mathrm{PII}_d(\xi, \Omega, \nu)$ respectively.

A convenient stochastic representation for Y is

$$Y = \xi + RL^{\mathrm{T}}S \tag{11}$$

where $L^{\mathrm{T}}L = \Omega$, the random vector S is uniformly distributed on the unit sphere in $\mathbb{R}^d$ and R is a positive scalar random variable independent of S, called the generating variate. An immediate consequence of this representation is that

$$(Y-\xi)^{\mathrm{T}}\Omega^{-1}(Y-\xi) \overset{\mathrm{d}}{=} R^2,$$

where $=^{\mathrm{d}}$ means equality in distribution.

Elliptical distributions are closed under affine transformations and conditioning. In particular they are closed under marginalization, in the following sense: consider the block partition $Y^{\mathrm{T}} = (Y_1^{\mathrm{T}}, Y_2^{\mathrm{T}})$ where $Y_1 \in \mathbb{R}^h$ and a corresponding partition for ξ and Ω; then

$$Y_1 \sim \mathrm{Ell}_h(\xi_1, \Omega_{11}, \tilde{f}_1).$$

Similarly, for the conditional density we have

$$(Y_1|Y_2 = y_2) \sim \mathrm{Ell}_h\{\xi_1 + \Omega_{12}\Omega_{22}^{-1}(y_2-\xi_2), \Omega_{11} - \Omega_{12}\Omega_{22}^{-1}\Omega_{21}, \tilde{f}^{Q_y}\},$$

where $Q_y = y_2^{\mathrm{T}}\Omega_{22}^{-1}y_2$. The density generators $\tilde{f}_1$ and $\tilde{f}^{Q_y}$ are not necessarily of the same form as $\tilde{f}$. Kano (1994) has shown that the form of the density generator is preserved under marginalization only in the case of elliptical distributions which can be obtained from a scale mixture of normal variates. This property is true, for instance, for multivariate Pearson type VII and II distributions. The generator $\tilde{f}^{Q_y}$ of the conditional distribution depends in general on the quantity Q_y, with the exception of the normal distribution.

2. Central symmetry and distributions obtained by its perturbation

Our starting-point is the following proposition, which is closely connected to lemma 1 of Azzalini and Capitanio (1999). Strictly speaking, the present statement is a little more restricted than the earlier result, but it has the major advantage of requiring a set of conditions whose fulfilment is far simpler to check, and still it represents a very general formulation.

The result refers to central symmetry, a simple and wide concept of symmetry, which is commonly in use in nonparametric statistics; see Zuo and Serfling (2000). Other researchers refer to the same property with alternative terms. A d-dimensional random variable Y is said to be centrally symmetric around a point ξ if $Y - \xi =^{\mathrm{d}} \xi - Y$. Since we shall deal with continuous variables, the above requirement implies that the corresponding density function f satisfies $f(y-\xi) = f(\xi-y)$ for all $y \in \mathbb{R}^d$, up to a negligible set. It is immediate to see that the condition of central symmetry is satisfied by various ample families; the elliptical densities are an important example.

Proposition 1. Denote by $f(y)$ the density function of a d-dimensional continuous random variable which is centrally symmetric around 0, and by G a scalar distribution function such that $G(-x) = 1 - G(x)$ for all real x. If $w(y)$ is a function from $\mathbb{R}^d$ to $\mathbb{R}$ such that $w(-y) = -w(y)$ for all $y \in \mathbb{R}^d$, then

$$2\, f(y)\, G\{w(y)\} \tag{12}$$

is a density function.

Proof. Denote by Y a random variable with density f, and by X a random variable with distribution function G, independent of Y. To show that $W = w(Y)$ has a distribution that is symmetric about 0, denote by A a Borel set of the real line and by $-A$ its mirror set obtained by reversing the sign of each element of A. Then, taking into account that Y and $-Y$ have the same distribution,

$$\mathbb{P}(W \in -A) = \mathbb{P}(-W \in A) = \mathbb{P}\{w(-Y) \in A\} = \mathbb{P}\{w(Y) \in A\},$$

showing that W has the property indicated. Then, on noting that $X - W$ has a distribution that is symmetric about 0, write

$$\tfrac{1}{2} = \mathbb{P}(X \leqslant W) = \mathbb{E}_Y[\mathbb{P}\{X \leqslant w(Y) | Y = y\}] = \int_{\mathbb{R}^d} G\{w(y)\}\, f(y)\, \mathrm{d}y,$$

which completes the proof.

To demonstrate graphically the ample flexibility that is attained by expression (12) for appropriate choices of f, G and w, we present the following example in the case $d = 2$. Consider the non-elliptical distribution

$$f(y) = \frac{(1 - y_1^2)^{a-1} (1 - y_2^2)^{b-1}}{4^{a+b-1}\, B(a, a)\, B(b, b)}, \qquad y = (y_1, y_2) \in (-1, 1)^2,$$

obtained by multiplication of two symmetric beta densities rescaled to the interval $(-1, 1)$, with positive parameters a and b. We perturb this density by choosing

$$G(x) = \frac{\exp(x)}{1 + \exp(x)},$$
$$w(y) = \frac{\sin(p_1 y_1 + p_2 y_2)}{1 + \cos(q_1 y_1 + q_2 y_2)}$$

where p_1, p_2, q_1 and q_2 are additional parameters. We have generated several plots of this type of density, obtaining an extremely rich set of surfaces, as indicated by the small collection of such densities given in Fig. 1. The plots indicate that the effect of perturbing f via expression (12) is far more complex than the effect introduced, say, by the skewing factor of the normal density (2). Clearly, the purpose of Fig. 1 is purely illustrative, and it is not proposed to use this class of density functions in practice without further investigation.

For a random variable with density (12), the stochastic representation given by Azzalini and Capitanio (1999), page 599, for a slightly different case is still valid. In fact, the conditions required there for its validity are actually those of proposition 1. Specifically, if Y has density function f and X is an independent variable with distribution function G, then

$$Z = \begin{cases} Y, & \text{if } X < w(Y), \\ -Y, & \text{if } X > w(Y) \end{cases} \tag{13}$$

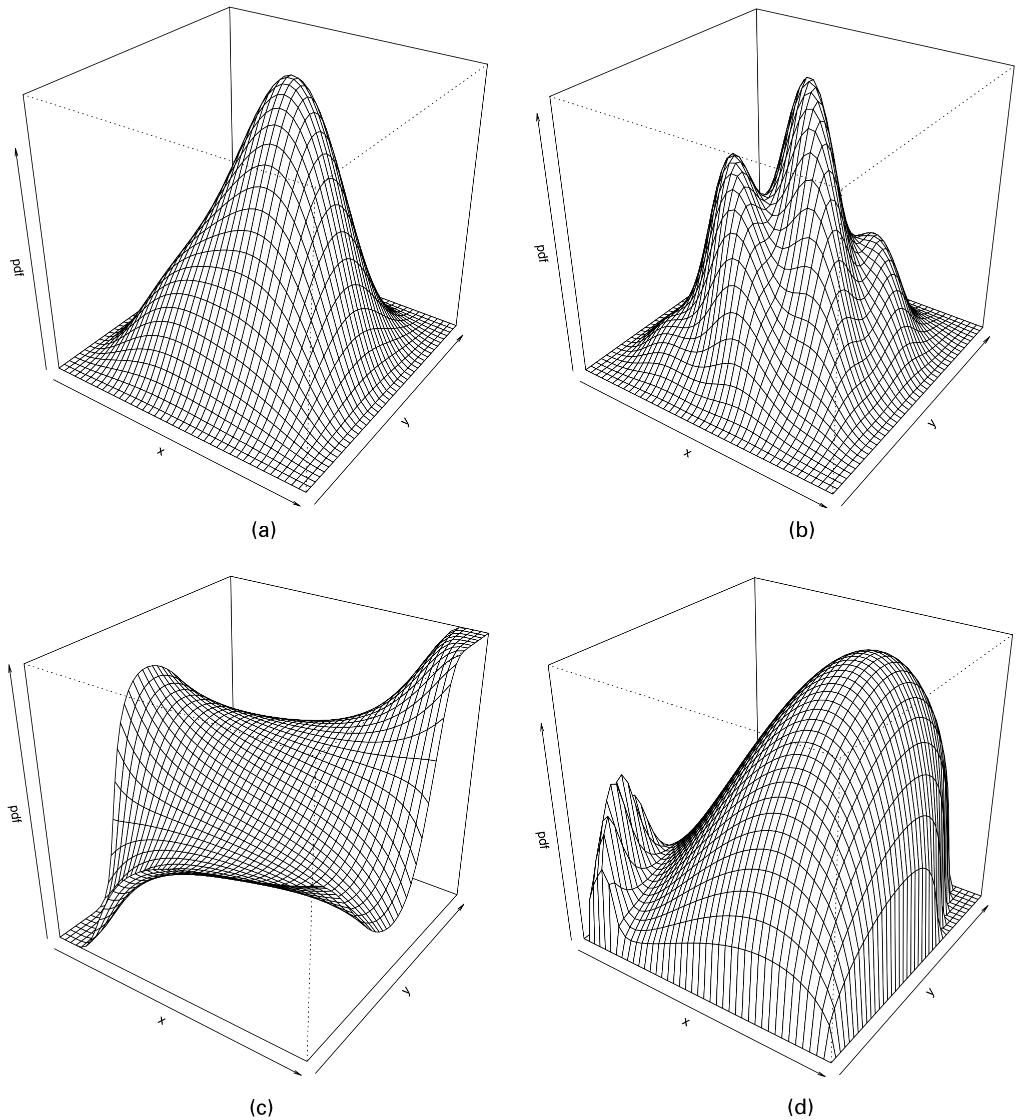

Fig. 1. Examples of perturbed symmetry beta densities: (a) $(a,b,p,q) = (2,3,(3,3),(0,0))$; (b) $(a,b,p,q) = (2,3,(8,8),(0,0))$; (c) $(a,b,p,q) = (3,1,(-1,3),(2,1))$; (d) $(a,b,p,q) = (3,1.5,(3.1),(2.5,1))$

has density function (12). Clearly, this provides an algorithm for generating Z and it will also turn out to be useful for theoretical purposes.

It can be shown that the conditioning method for generating SN random variables from expression (5) is a special case of expression (13). In fact, from a consideration of the residual part of U_0 after removing the regression on U, define the variable

$$\tilde{X} = -(1 - \delta^{\mathrm{T}} \bar{\Omega}^{-1} \delta)^{-1/2} (U_0 - \delta^{\mathrm{T}} \bar{\Omega}^{-1} U) \sim N(0,1), \tag{14}$$

independent of U. After substituting symbols, the condition $\tilde{X} < \alpha^{\mathrm{T}} U$ of the first of equations (13) is equivalent to $U_0 > 0$, if α is given by equation (4); an analogous correspondence exists for the second of equations (13). Similarly, the stochastic representation of a variate with density (10) via $(U|U_0 + \tau > 0)$ could be reformulated in terms of the condition $\tilde{X} < \alpha_0 + \alpha^{\mathrm{T}} U$.

It is immediate that, if f is an elliptical density, G corresponds to a distribution that is symmetric about 0 and $w(y) = \alpha^{\mathrm{T}} y$ for some $\alpha \in \mathbb{R}^d$, then the conditions required by proposition 1 are fulfilled. We then obtain the family of densities that is produced by corollary 2 of Azzalini and Capitanio (1999).

Proposition 2. Denote by Y and Z two d-dimensional random variates having density function f and expression (12) respectively, satisfying the conditions of proposition 1. If $t(\cdot)$ is a function from $\mathbb{R}^d$ to some Euclidean space, such that $t(-y) = t(y)$ for all $y \in \mathbb{R}^d$, then

$$t(Y) \stackrel{\mathrm{d}}{=} t(Z).$$

Proof. The proof is immediate from representation (13).

A key example of this result is obtained when $t(y)$ represents the distance from the origin. Since any choice of $t(\cdot)$ must satisfy the symmetry condition $t(y) = t(-y)$, then the probability distribution of the distance of a random point from the origin is the same for Y and for Z. In particular we can write

$$Y^{\mathrm{T}} B Y \stackrel{\mathrm{d}}{=} Z^{\mathrm{T}} B Z$$

for any positive definite matrix B. A result that is similar to proposition 2 for the case when f is an elliptical distribution has been given by Genton and Loperfido (2002).

A related set of applications of proposition 2 is offered by various results on quadratic forms of SN variates, all of which lead to the conclusion that known distributional results for normal variates still hold if the variates are of SN type. This set of results includes propositions 7–9 of Azzalini and Capitanio (1999) and propositions 1, 2 and 6 (parts 1 and 3) of Loperfido (2001). For these conclusions, we must consider functions $t(\cdot)$ in proposition 2 taking on values in an appropriate Euclidean space, e.g. $\mathbb{R}^+ \times \mathbb{R}^+$ if the independence of two quadratic forms is under consideration. Note that propositions 8 and 9 of Azzalini and Capitanio (1999) have added conditions on the α-parameter, but these are not necessary. There is no conflict with the present conclusions since in their proposition 8 this extra condition is part of a sufficiency requirement, and their proposition 9 (a Fisher–Cochran type of theorem) was stated in a more restricted form than is actually possible.

We conclude this section with a discussion on possible generalizations of proposition 1. A very general form of density resembling density (12) is along the following lines. Denote by $X = (X_1, \ldots, X_m)^{\mathrm{T}}$ an m-dimensional random variable with distribution function G, by Y an independent d-dimensional random variable with density function f and by $w_1(y), \ldots, w_m(y)$ a set of functions from $\mathbb{R}^d$ to $\mathbb{R}$. For the moment, we remove any assumptions on f, G and the w_is; there is no loss of generality in assuming that $w_i(0) = 0$. Then

$$p^{-1} G\{w_1(y) + b_1, \ldots, w_m(y) + b_m\} f(y) \tag{15}$$

is a density function for any choice of the real numbers $b_1, \ldots, b_m$, if

$$p = \mathbb{P}\{X_1 - w_1(Y) \leqslant b_1, \ldots, X_m - w_m(Y) \leqslant b_m\}.$$

Clearly, the difficulty is in computing the normalizing constant p. This task is amenable when X and Y are multivariate normal variables. A rather simple special case of expression (15) is given by expression (10) where G is the scalar normal distribution function, and f is $\phi_d(x; \Omega)$. An instance of density (15) with multivariate G is given by Sahu *et al.* (2001); in their case, f is the d-dimensional normal density, G is the d-dimensional normal distribution function, the w_js are

d linear combinations of y and all b_js are 0. The multivariate distribution sketched by Azzalini (1985), section 4, and the multiple-constraint model outlined by Arnold and Beaver (2000a), section 6, has a G which is the product of m ($m \geqslant 1$) terms of type $\Phi(\alpha_i y_i)$ or $\Phi(\alpha_i^{\mathrm{T}} y + b_i)$ respectively. The 'general multivariate skew normal distribution' mentioned by Gupta *et al.* (2001), section 5, is even more general since they adopted a G which is the m-dimensional normal distribution function.

When f or G or both, in expression (15), are not of Gaussian type, the evaluation of p is generally much more problematic. Some form of restrictions must be imposed, however, not only to make the problem tractable but also because it has little meaning to consider density (15) in its full generality which is so broad as to lose nearly any structure. A reasonable setting is as follows: suppose that f and G are both centrally symmetric and $w_i(-y) = -w_i(y)$ for all $y \in \mathbb{R}^d$. Then, by using essentially the same argument as in the proof of proposition 1, we conclude that $W = (W_1, \ldots, W_m) = (w_1(Y), \ldots, w_m(Y))$ is centrally symmetric; therefore so is $V = (X_1 - W_1, \ldots, X_m - W_m)$, by using the properties of centrally symmetric functions. A tractable instance of this setting is offered by the skew Cauchy distribution and its variants discussed by Arnold and Beaver (2000b), using a univariate G.

3. Skew elliptical densities

This section focuses on an important subclass of density (12) with the component f of elliptical form, aiming at three main goals. The first is to prove that the two forms of skew elliptical densities that were introduced by Azzalini and Capitanio (1999), page 599, and by Branco and Dey (2001) are closely connected. The second goal is to show that the relationships between the three forms of stochastic representation of an SN variate recalled in Section 1.2 carry over to skew elliptical variates. Furthermore, an analogue of stochastic representation (11) for elliptical variates is obtained for skew elliptical variates.

3.1. Skew elliptical densities by conditioning

For simplicity of presentation, we shall work with correlation matrices and location parameter 0. For the rest of this section, U^* denotes a $(d+1)$-dimensional variate partitioned into a scalar component U_0 and a d-dimensional vector U.

Branco and Dey (2001) introduced a class of skew elliptical distributions generated by applying to a $(d+1)$-dimensional elliptical variate the same conditioning method as described in Section 1.2 in connection with the SN distribution. The following proposition recalls their key statement, up to some inessential changes of notation.

Proposition 3. Consider the random vector $U^* \sim \mathrm{Ell}_{d+1}(0, \Omega^*, \tilde{f})$ where Ω^* is defined in expression (5). Then the probability density function of $Z = (U|U_0 > 0)$ is

$$2\, f_U(z; \bar{\Omega}) \int_{-\infty}^{\alpha^{\mathrm{T}} z} c_1 \tilde{f}^{Q_z}(y^2)\, \mathrm{d}y \tag{16}$$

where

$$Q_z = z^{\mathrm{T}} \bar{\Omega}^{-1} z, \tag{17}$$

the vector α is defined in equation (4), f_U is the density of U, $\tilde{f}^{Q_z}(\cdot)$ is the density generator of $(U_0|U = z)$ and c_1 is the associated normalizing constant.

For later use, note that an alternative expression for density (16) is

$$2\int_0^\infty c_{d+1}\,\tilde{f}(u^{*\mathrm{T}}\Omega^{*-1}u^*)|\Omega^*|^{-1/2}\,\mathrm{d}u_0. \tag{18}$$

On defining $F^{Q_z}(\cdot)$ to be the distribution function corresponding to the density generator $\tilde{f}^{Q_z}(\cdot)$, this result led Branco and Dey (2001) to rewrite expression (16) in the form

$$2\,f_U(z;\bar{\Omega})\,F^{Q_z}(\alpha^{\mathrm{T}}z) \tag{19}$$

where the distribution function F^{Q_z} is actually varying at each selected point z. This expression appears to be different from expression (12) where a fixed distribution function F is involved.

However, when the quantity Q_z can be removed from the argument of the integral in expression (16) by means of a suitable change in variable, the resulting density function will become

$$2\,f_U(z;\bar{\Omega})\,F\{w(z)\} \tag{20}$$

where F is a univariate distribution function and w is such that $w(z)=h(\alpha^{\mathrm{T}}z, z^{\mathrm{T}}\bar{\Omega}^{-1}z)$ for some function h from $\mathbb{R}\times\mathbb{R}^+$ to $\mathbb{R}$. It is easy to show that the property $w(-z)=-w(z)$ must hold; hence density (20) is of type (12).

It is difficult to state general conditions under which a density of type (19) can be transformed into one of form (20), but special cases where this is feasible exist. We shall now examine in detail two important cases of this form, namely when U^* has either a PVII_{d+1} or a PII_{d+1} distribution, which are among those considered by Branco and Dey (2001).

Proposition 4. If the random vector U^* has a $\mathrm{PVII}_{d+1}(0,\Omega^*,M,\nu)$ distribution, then the probability density function of $Z=(U|U_0>0)$ is

$$2\,f_U(z;\bar{\Omega})\,F_1\{\alpha^{\mathrm{T}}z(\nu+Q_z)^{-1/2};M,1\},\qquad z\in\mathbb{R}^d, \tag{21}$$

where Q_z is given by equation (17), f_U is the density of a $\mathrm{PVII}_d\big(0,\bar{\Omega},M-\frac{1}{2},\nu\big)$ distribution and $F_1(\cdot;M,1)$ is the cumulative probability function of a $\mathrm{PVII}_1(0,\bar{\Omega},M,1)$ distribution.

Proof. Using results in Fang *et al.* (1990), pages 82–83, we have

$$c_1\,\tilde{f}^{Q_z}(y^2)=\frac{\Gamma(M)}{\pi^{1/2}\,\Gamma(M-\frac{1}{2})}(\nu+Q_z)^{-1/2}\left(1+\frac{y^2}{\nu+Q_z}\right)^{-M}$$

and

$$f_U(z;\bar{\Omega})=\frac{\Gamma(M-\frac{1}{2})}{|\bar{\Omega}|^{1/2}(\pi\nu)^{d/2}\,\Gamma\{M-(d+1)/2\}}\left(1+\frac{Q_z}{\nu}\right)^{-M+1/2},$$

i.e. the densities of a $\mathrm{PVII}_1(0,1,M,\nu+Q_z)$ and of a $\mathrm{PVII}_d(0,\bar{\Omega},M-\frac{1}{2},\nu)$ variate with parameters $M-\frac{1}{2}$ and ν respectively. On setting $x=y(\nu+Q_z)^{-1/2}$, the integral in expression (16) becomes

$$\int_{-\infty}^{\alpha^{\mathrm{T}}z(\nu+Q_z)^{-1/2}}\frac{\Gamma(M)}{\pi^{1/2}\,\Gamma(M-\frac{1}{2})}(1+x^2)^{-M}\,\mathrm{d}x$$

which is the distribution function of a $\mathrm{PVII}_1(0,1,M,1)$ variate evaluated at the point $\alpha^{\mathrm{T}}z(\nu+Q_z)^{-1/2}$.

3.1.1. Example 1: skew t-distribution
The relevance of the PVII_d class is due to the inclusion of the multivariate t-family as the special case when $M = (d+\nu)/2$. The corresponding specification of proposition 4 produces then a form of multivariate skew t-density. Since Section 4 will be entirely dedicated to this distribution, we defer a detailed discussion until then.

Proposition 5. If the $(d+1)$-dimensional elliptical random vector U^* has a $\mathrm{PII}_{d+1}(0, \Omega^*, \nu)$ distribution, then the probability density function of $Z = (U|U_0 > 0)$ is

$$2\, f_U(z; \bar{\Omega})\, F_1\{\alpha^{\mathrm{T}} z (1 - Q_z)^{-1/2}; \nu\}, \qquad z \in (-1, 1)^d, \tag{22}$$

where Q_z is given by equation (17), f_U is the density of a $\mathrm{PII}_d(0, \bar{\Omega}, \nu + \frac{1}{2})$ variate and $F_1(\cdot; \nu)$ is the distribution function of a $\mathrm{PII}_1(0, 1, \nu)$ variate.

Proof. The proof is identical with that of proposition 4, considering the densities of marginal and conditional distributions of PII, as defined in Fang *et al.* (1990), pages 89–91.

The absence of Q_z in the conditional density characterizes the multivariate normal distribution among the members of the elliptical family. This fact can be used to produce an analogous characterization of the SN distribution within the skew elliptical family.

Proposition 6. The function w in expression (20) is such that $w(z) = \alpha^{\mathrm{T}} z$ if and only if U^* is Gaussian, i.e. Z is SN.

Proof. The density of $(U|U_0 = z)$ does not depend on Q_z if and only if U^* is Gaussian; see theorem 4.12 of Fang *et al.* (1990). In this case, the integral in expression (16) becomes $\Phi(\alpha^{\mathrm{T}} z)$, so $Z \sim \mathrm{SN}_d(0, \bar{\Omega}, \alpha)$.

Several parallels between the SN distribution and other types of skew elliptical distributions have already been shown. The next result allows us to construct a random variable $\tilde{X}$ playing a role analogous to the one in equation (14) for the skew version of a PVII_d and PII_d distribution.

Proposition 7. Let $U^* \sim \mathrm{PVII}_{d+1}(0, \Omega^*, M, \nu)$. Then

$$\tilde{X} = -(1 - \delta \bar{\Omega}^{-1} \delta)^{-1/2} (U_0 - \delta^{\mathrm{T}} \bar{\Omega}^{-1} U)(\nu + U^{\mathrm{T}} \bar{\Omega}^{-1} U)^{-1/2} \sim \mathrm{PVII}_1(0, 1, M, 1),$$

independent of U. If $U^* \sim \mathrm{PII}_{d+1}(0, \Omega^*, \nu)$ then

$$\tilde{X} = -(1 - \delta \bar{\Omega}^{-1} \delta)^{-1/2} (U_0 - \delta^{\mathrm{T}} \bar{\Omega}^{-1} U)(1 - U^{\mathrm{T}} \bar{\Omega}^{-1} U)^{-1/2} \sim \mathrm{PII}_1(0, 1, \nu),$$

independent of U.

Proof. The proof is by direct calculation.

Therefore, we can set

$$Z = \begin{cases} U & \text{if } \tilde{X} < w(U), \\ -U & \text{if } \tilde{X} > w(U), \end{cases}$$

where $w(z)$ is the transformation of z used in the argument of F_1 in expressions (21) and (22); it is intended that the appropriate distribution of U^* and transformation $\tilde{X}$ have been selected. This formula establishes a method of type (13) to generate a skew PVII_d and a skew PII_d variate.

The connections between the proposal of Azzalini and Capitanio (1999) and that of Branco and Dey (2001) can be summarized as follows. The conditioning argument which is one of the mechanisms to generate the SN distribution from the normal distribution can be adopted to generate a form of skew elliptical distributions from the elliptical ones, leading to density (19), or some similar form as obtained by Branco and Dey. This type of expression can, at least in some important special cases, be transformed into one where the skewing factor of f is a fixed distribution function, as shown by expressions (21) and (22). These expressions are of type (12), which is essentially the form of Azzalini and Capitanio. The natural question is whether all densities of type (19) can be rewritten in the form (12), but we have been unable to prove this fact in general. Note that the converse inclusion is not true, i.e. not all densities of type (12) can be written in the form (19), unless additional restrictions are imposed on the components of density (12), besides the obvious condition that f is elliptical.

The next result concerns a stochastic representation of type (11) for distributions of type (12) when the density f is elliptical. For example, this representation is valid for the skew elliptical densities defined in Azzalini and Capitanio (1999), page 599, and for the skew versions of PVII_d and PII_d examined earlier.

Proposition 8. If Z has a density of type (12), where f is the density of $U \sim \mathrm{Ell}_d(\xi, \bar{\Omega}, \tilde{f})$, then Z admits the stochastic representation

$$Z = \xi + RL^{\mathrm{T}} S' \tag{23}$$

where $\bar{\Omega} = L^{\mathrm{T}} L$, $R > 0$, has the same distribution as the radius of the stochastic representation (11) of U, and S' has a non-uniform distribution on the unit sphere of $\mathbb{R}^d$. Specifically, using spherical co-ordinates, the density of S' is equal to

$$\frac{\Gamma(d/2)}{\pi^{d/2}} \prod_{k=1}^{d-2} \sin(\theta_k)^{d-k-1} \, \mathbb{P}\{X \leqslant w_L^*(\theta_1, \ldots, \theta_{d-1}, R)\},$$

where $w_L^*(\cdot)$ is a suitable function from $\mathbb{R}^d$ to $\mathbb{R}$, and X is an independent random variable having distribution function G. Furthermore, the conditional distribution of S' given $R = r$ is of type (12), with density

$$\frac{\Gamma(d/2)}{\pi^{d/2}} \prod_{k=1}^{d-2} \sin(\theta_k)^{d-k-1} \, G\{w_L^*(\theta_1, \ldots, \theta_{d-1}, r)\}.$$

Proof. For a proof see appendix A of the full version of the paper, which includes also a definition of the function $w_L^*(\cdot)$.

3.1.2. Example 2: stochastic representation (23) for skew normal distribution

If $Z \sim \mathrm{SN}_d(\xi, \bar{\Omega}, \alpha)$, then by applying proposition 8 we obtain $R^2 \sim \chi_d^2$ and the following spherical co-ordinates representation of the marginal distribution of S':

$$\begin{aligned} f_\theta(\theta) = {} & 2\frac{\Gamma(d/2)}{2\pi^{d/2}} \prod_{k=1}^{d-2} \sin(\theta_k)^{d-k-1} \\ & \times \mathbb{P}[X \leqslant R\{\alpha_1^* \cos(\theta_1) + \alpha_2^* \sin(\theta_1) \cos(\theta_2) + \ldots + \alpha_d^* \sin(\theta_1) \ldots \sin(\theta_{d-1})\}], \end{aligned}$$

where $\theta = (\theta_1, \theta_2, \ldots, \theta_{d-1})^{\mathrm{T}}$, $\alpha^* = L\alpha$ and $X \sim N(0, 1)$ is independent of R. Finally, noting that $d^{1/2} X R^{-1}$ has a t-distribution with d degrees of freedom, we have

$$\begin{aligned} f_\theta(\theta) = {} & 2\frac{\Gamma(d/2)}{2\pi^{d/2}} \prod_{k=1}^{d-2} \sin(\theta_k)^{d-k-1} \\ & \times T_1[d^{1/2}\{\alpha_1^* \cos(\theta_1) + \alpha_2^* \sin(\theta_1) \cos(\theta_2) + \ldots + \alpha_d^* \sin(\theta_1) \ldots \sin(\theta_{d-1})\} d] \end{aligned}$$

where $T_1(\cdot; d)$ is the distribution function of a scalar t-distribution with d degrees of freedom.

3.2. *Skew elliptical densities by transformation method*

The next result shows how the class of skew elliptical distributions mirrors another property of the SN distribution. In fact the class of skew elliptical densities that is obtained via the conditioning method is equivalent to that obtained by applying the transformation method recalled in Section 1.2.

Proposition 9. Consider the random vector $(U_0, U) \sim \mathrm{Ell}_{d+1}(0, \Psi^*, \tilde{f})$ where Ψ^* is as in expression (6), and define

$$Z_j = \delta_j |U_0| + (1 - \delta_j^2)^{1/2} U_j, \qquad j = 1, \ldots, d, \tag{24}$$

where $-1 < \delta_j < 1$. Then the density of $(Z_1, \ldots, Z_d)$ is of type (16), where

$$\begin{aligned} \lambda_i &= \delta_i (1 - \delta_i^2)^{-1/2}, \qquad i = 1, \ldots, d, \\ \Delta &= \mathrm{diag}\{(1 + \lambda_1^2)^{-1/2}, \ldots, (1 + \lambda_d^2)^{-1/2}\}, \\ \Omega &= \Delta(\Psi + \lambda\lambda^{\mathrm{T}})\Delta, \\ \alpha &= (1 + \lambda^{\mathrm{T}}\Psi\lambda)^{-1/2} \Delta^{-1} \Psi^{-1} \lambda. \end{aligned}$$

Proof. First note that the joint density function of $|U_0|$ and U takes the form $2c_{d+1}\tilde{f}(\cdot)$. Denote by B the $(d+1) \times (d+1)$ matrix that is implicitly defined by equation (24) such that $(Z_0, Z_1, \ldots, Z_d)^{\mathrm{T}} = B(|U_0|, U^{\mathrm{T}})^{\mathrm{T}}$, and apply the usual formulae for linear transforms. Then the density function of $(Z_1, \ldots, Z_d)$ turns out to be

$$2 \int_0^\infty c_{d+1} \tilde{f}\{(z_0, z^{\mathrm{T}}) A^{-1} (z_0, z^{\mathrm{T}})^{\mathrm{T}}\} |A|^{-1/2} \, \mathrm{d}x_0$$

where $A = B\Psi^* B^{\mathrm{T}}$ is a correlation matrix. Taking into account expression (18) the result follows.

An immediate consequence of the transformation method is a further generating method for the bivariate case. Again, this reproduces for the skew elliptical family a generation method that is known to hold for the SN distributions.

Proposition 10. If $(U_0, U) \sim \mathrm{Ell}_2(0, \Omega^*, \tilde{f})$, the class generated by $Z = \max(U_0, U)$ is equal to the class generated by the transformation method of proposition 9 with $d = 2$.

Proof. First note that $\max(U_0, U) = \frac{1}{2}|U - U_0| + \frac{1}{2}(U + U_0)$. As the joint distribution of $(U - U_0)(2 - 2\rho)^{-1/2}$ and $(U + U_0)(2 + 2\rho)^{-1/2}$ is $\mathrm{Ell}_2(0, I, \tilde{f})$, where ρ denotes the off-diagonal elements of Ω^*, the result follows by direct application of proposition 9 on imposing $\delta = \{\frac{1}{2}(1 - \rho)\}^{1/2}$.

4. A skew *t*-distribution

For the rest of the paper we shall focus on the development of an asymmetric version of the multivariate Student *t*-distribution, already sketched in Section 3.1. The purpose of the present section is to provide additional support for its definition and to examine more closely its properties. Connected inferential aspects will be discussed in the subsequent section.

4.1. *Definition and density*

The usual construction of the *t*-distribution is via the ratio of a normal variate and an appropriate transformation of a χ^2-variate. If we want to introduce an asymmetric variant of the *t*-distribution, quite a natural option is to replace the normal variate above by an SN variate.

A preliminary result on gamma variates is required. In our parameterization, gamma(ψ, λ) refers to a gamma variable with mean value ψ/λ and variance ψ/λ^2.

Lemma 1. If $V \sim \text{gamma}(\psi, \lambda)$, then for any $a, b \in \mathbb{R}$

$$\mathbb{E}\{\Phi(a\surd V + b)\} = \mathbb{P}\{T \leqslant a\surd(\psi/\lambda)\}$$

where T denotes a non-central t-variate with 2ψ degrees of freedom and non-centrality parameter $-b$.

Proof. Let $U \sim N(0, 1)$; then

$$\begin{aligned}\mathbb{E}\{\Phi(a\surd V + b)\} &= \mathbb{E}_V\{\mathbb{P}(U \leqslant a\surd v + b | V = v)\} \\ &= \mathbb{E}_V\left[\mathbb{P}\left\{\frac{U-b}{(v\lambda/\psi)^{1/2}} \leqslant a\left(\frac{\psi}{\lambda}\right)^{1/2} \middle| V = v\right\}\right] \\ &= \mathbb{P}\{T' \leqslant a(\psi/\lambda)^{1/2}\}\end{aligned}$$

where

$$T = \frac{U-b}{(V\lambda/\psi)^{1/2}}$$

has the quoted t-distribution.

As anticipated earlier, we define the skew t-distribution as the one corresponding to the transformation

$$Y = \xi + V^{-1/2} Z \tag{25}$$

where Z has density function (2) with $\xi = 0$, and $V \sim \chi^2_\nu/\nu$, independent of Z. An equivalent interpretation of Y is to regard it as a scale mixture of SN variates, with mixing scale factor $V^{-1/2}$. Application of lemma 1 to a gamma$(\frac{1}{2}\nu, \frac{1}{2}\nu)$ variate and some simple algebra lead to the density of Y, which is

$$f_Y(y) = 2 t_d(y; \nu)\, T_1\left\{\alpha^{\mathrm{T}}\omega^{-1}(y-\xi)\left(\frac{\nu+d}{Q_y+\nu}\right)^{1/2}; \nu+d\right\} \tag{26}$$

where ω is defined at the beginning of Section 1.2,

$$Q_y = (y-\xi)^{\mathrm{T}}\Omega^{-1}(y-\xi),$$

$$t_d(y; \nu) = \frac{1}{|\Omega|^{1/2}} g_d(Q_y; \nu) = \frac{\Gamma\{(\nu+d)/2\}}{|\Omega|^{1/2}(\pi\nu)^{d/2}\,\Gamma(\nu/2)}\left(1+\frac{Q_y}{\nu}\right)^{-(\nu+d)/2}$$

is the density function of a d-dimensional t-variate with ν degrees of freedom and $T_1(x; \nu+d)$ denotes the scalar t distribution function with $\nu + d$ degrees of freedom. We shall call distribution (26) skew t and write

$$Y \sim \mathrm{St}_d(\xi, \Omega, \alpha, \nu). \tag{27}$$

It is easy to check that density (26) coincides with that sketched in Section 3.1 using proposition 4, which is of type (12). Moreover, for the reasons explained in Section 3.1, distribution (26) coincides in turn with the skew t-distribution of Branco and Dey (2001), although this equality is not visible from their derivation because they did not provide the above closed form expression of the density.

Therefore, we have seen that several different ways to define a skew t-distribution all lead to the same density (26). Although additional proposals to introduce a form of a skew t-density are possible, this one has the advantage of arising from various generating criteria, which in turn are linked to other portions of literature.

A reviewer of this paper remarked that, if we set $d = 1$, density (26) does not reduce to the form $2\,t_1(y;\nu)\,T_1(\alpha y;\nu)$, which seems to be the 'most natural' univariate form of skew t-density generated by lemma 1 of Azzalini (1985), a forerunner of proposition 1. Although the latter density has the appeal of a slightly simpler mathematical expression, the arguments indicated in the previous paragraph lead us to prefer density (26). In fact, one could reverse the reasoning and claim that lemma 1 of Azzalini (1985) 'should' have been stated in the form of proposition 1 for $d = 1$; in other words, there is no reason to restrict $w(y)$ to the linear form αy, especially outside the normal case.

Alternative proposals of univariate skew t-distributions have been made by Fernández and Steel (1998), constructed similarly to the so-called two-piece normal density, and by Jones (2001), developed by Jones and Faddy (2003), which is based on a suitable transformation of a beta density. A multivariate form of skew t-distribution has been proposed by Jones (2002) but the associated inferential aspects have not been discussed. The alternative form of multivariate skew t-distribution that was considered by Sahu *et al.* (2001) coincides with equation (26) in the case $d = 1$; for general d, their density involves the multivariate t-distribution function. The density that is examined in this paper allows a relatively simple mathematical treatment, and it is more naturally linked to the SN distribution, via mechanisms that have already been mentioned. As a consequence, the distribution enjoys various useful formal properties, which will be examined in the remaining part of this section.

4.2. Some properties

4.2.1. Distribution function

For simplicity of exposition, we obtain the distribution function of Y in the 'standard' case with $\xi = 0$ and $\Omega = \bar{\Omega}$. Bearing in mind the representation of Z based on conditioning, write

$$\mathbb{P}(Y \leqslant y) = 2\,\mathbb{P}\left\{V^{-1/2}\begin{pmatrix}-U_0\\ U\end{pmatrix} \leqslant \begin{pmatrix}0\\ y\end{pmatrix}\right\} = 2\,\mathbb{P}\left\{T' \leqslant \begin{pmatrix}0\\ y\end{pmatrix}\right\}$$

where (U_0, U) has distribution (5) and the inequality signs are intended componentwise. The last expression involves the integral of a multivariate $(d+1)$-dimensional t-variate T' with dispersion matrix similar to that of distribution (5), but with reversed sign of δ. Algorithms for computing this type of distribution function are given by Genz and Bretz (1999).

4.2.2. Moments

Using the representation (25), it is easy to compute the moments of Y. For algebraic convenience, we assume $\xi = 0$ throughout. If $\mathbb{E}(Y^{(m)})$ denotes a moment of order m, write

$$\mathbb{E}(Y^{(m)}) = \mathbb{E}(V^{-m/2})\mathbb{E}(Z^{(m)}) \tag{28}$$

where Z has density function (2) with $\xi = 0$. It is well known that

$$\mathbb{E}(V^{-m/2}) = \frac{(\nu/2)^{m/2}\,\Gamma\{\frac{1}{2}(\nu - m)\}}{\Gamma(\frac{1}{2}\nu)},$$

whereas, for the expressions of $\mathbb{E}(Z^{(m)})$, we use results given by Azzalini and Capitanio (1999) and by Genton *et al.* (2001).

First, we apply equation (28) to the scalar case. On defining

$$\mu = \delta\left(\frac{\nu}{\pi}\right)^{1/2} \frac{\Gamma\{\frac{1}{2}(\nu-1)\}}{\Gamma(\frac{1}{2}\nu)}, \qquad \nu > 1, \tag{29}$$

we obtain, for $\xi = 0$,

$$\mathbb{E}(Y) = \omega\mu,$$
$$\operatorname{var}(Y) = \omega^2\left(\frac{\nu}{\nu-2} - \mu^2\right),$$

provided that ν is larger than the corresponding order of the moment. These expressions are in agreement with those given by Branco and Dey (2001), but higher moments were not given there. Further application of equation (28) gives the third- and fourth-order moments, leading to the indices of skewness and kurtosis

$$\gamma_1 = \mu\left\{\frac{\nu(3-\delta^2)}{\nu-3} - \frac{3\nu}{\nu-2} + 2\mu^2\right\}\left(\frac{\nu}{\nu-2} - \mu^2\right)^{-3/2}, \qquad \text{if } \nu > 3,$$

$$\gamma_2 = \left\{\frac{3\nu^2}{(\nu-2)(\nu-4)} - \frac{4\mu^2\nu(3-\delta^2)}{\nu-3} + \frac{6\mu^2\nu}{\nu-2} - 3\mu^4\right\}\left(\frac{\nu}{\nu-2} - \mu^2\right)^{-2} - 3, \qquad \text{if } \nu > 4.$$

In the multivariate case, we obtain from equation (28) that $\mathbb{E}(Y) = \omega\mu$ still holds, provided that $\nu > 1$ and equation (29) and ω are intended in vector and matrix form respectively; furthermore

$$\mathbb{E}(YY^{\mathrm{T}}) = \frac{\nu}{\nu-2}\Omega, \qquad \text{if } \nu > 2,$$

leading to

$$\operatorname{var}(Y) = \frac{\nu}{\nu-2}\Omega - \omega\mu\mu^{\mathrm{T}}\omega.$$

4.2.3. Linear and quadratic forms

Consider the affine transformation $a + AY$ where $a \in \mathbb{R}^m$ and A is an $m \times d$ constant matrix of rank m. Using equation (25) we can write

$$a + AY = \xi' + V^{-1/2}AZ$$

where $\xi' = a + A\xi$. Take into account that

$$AZ \sim \mathrm{SN}_m(0, A\Omega A^{\mathrm{T}}, \alpha')$$

on the grounds of results given by Azzalini and Capitanio (1999) where the explicit expression for α' is given; similar results, but in a more convenient form, are provided by Capitanio *et al.* (2003), appendix A.2. Therefore we obtain

$$a + AY \sim \mathrm{St}_m(\xi', A\Omega A^{\mathrm{T}}, \alpha', \nu).$$

In particular for a single component, Y_r say ($r \in \{1, \ldots, d\}$), we have

$$Y_r \sim \mathrm{St}(\xi_r, \omega_{rr}, \alpha'_r, \nu)$$

where α'_r is given by expression (10) of Capitanio *et al.* (2003).

Similarly, for a quadratic form, $Q = (Y-\xi)^{\mathrm{T}} B (Y-\xi)$, where B is a symmetric $d \times d$ matrix, we can write

$$Q = Z^{\mathrm{T}} BZ/V.$$

For appropriate choices of B, the distribution of $Z^{\mathrm{T}} BZ$ is $\chi^2_{\nu'}$ for some value ν' of the degrees of freedom. One such case is expression (9), where $B = \Omega^{-1}$. Azzalini and Capitanio (1999), section 3.3, considered more general forms of B; see also Genton *et al.* (2001) for additional results. In all cases when the χ^2-property holds for Z, we can state immediately

$$Q/\nu' \sim F(\nu', \nu).$$

This property allows us to produce Healy-type plots (Healy, 1968) as a diagnostic tool in data fitting, similarly to the normal and SN case, just using the Snedecor distribution as the reference distribution instead of the χ^2-distribution. This device will be illustrated in the subsequent numerical work.

4.2.4. An extended skew t-distribution

If the component Z in equation (25) is taken to have distribution (10) rather than (2), we obtain a density which parallels the role of distribution (10) for skew t-densities; this is now discussed briefly.

By using lemma 1 again, the new density turns out to be of type (26), except that T_1 refers now to a t-distribution with non-centrality parameter $-\tau(1-\delta^{\mathrm{T}}\bar{\Omega}^{-1}\delta)^{-1/2}$ and the normalizing constant 2 is replaced by $1/\Phi(\tau)$. Additional properties are available in the extended version of the paper.

5. Statistical aspects of the skew t-distribution

5.1. Likelihood inference

Consider n independent observations satisfying a regression model of the type

$$y_i \sim \mathrm{St}_d(\xi_i, \Omega, \alpha, \nu), \qquad \xi_i = \beta^{\mathrm{T}} x_i,$$

for $i = 1, \ldots, n$; here x_i is a p-dimensional vector and β is a $p \times d$ matrix of parameters. Also let

$$X = (x_1, x_2, \ldots, x_n)^{\mathrm{T}}$$

be the $n \times p$ design matrix. We are effectively considering a multivariate regression model with error term of skew t type. It would be inappropriate to use such a distribution, and in fact even a regular elliptical distribution, for the joint modelling of the n observations, since usually these are supposed to behave independently.

It is convenient to reparameterize the problem by writing

$$\Omega^{-1} = A^{\mathrm{T}} \mathrm{diag}\{\exp(-2\rho)\}\, A = A^{\mathrm{T}} DA,$$
$$\eta = \omega^{-1}\alpha$$

where A is an upper triangular $d \times d$ matrix with diagonal terms equal to 1 and $\rho \in \mathbb{R}^d$. The contribution of individual i to the log-likelihood function for the parameter $\theta = (\beta, A, \rho, \eta, \log(\nu))$ is then

$$l_i(\theta) = \log(2) + \tfrac{1}{2}\log|D| + \log\{g_d(Q_i; \nu)\} + \log[T_1\,\{t(L_i, Q_i, \nu); \nu + d\}]$$

where

$$\begin{aligned} u_i &= y_i - \beta^{\mathrm{T}} x_i, \\ Q_i &= u_i^{\mathrm{T}} \Omega^{-1} u_i, \\ L_i &= \alpha^{\mathrm{T}} \omega^{-1} u_i, \\ t(L, Q, \nu) &= L \left(\frac{\nu + d}{Q + \nu} \right)^{1/2}. \end{aligned}$$

Maximization of the log-likelihood function must be accomplished numerically. To improve efficiency, the derivatives of the log-likelihood can be supplied to an optimization algorithm; details for computing these derivatives are given in an appendix of the extended version of the paper. A suite of R routines for evaluating the above log-likelihood and its derivatives has been developed, and it is available at the World Wide Web address indicated earlier.

In connection with the SN distribution, Azzalini (1985) and Azzalini and Capitanio (1999) have highlighted some problematic aspects of the likelihood function. A key feature is that the profile log-likelihood function for α always has a stationarity point at $\alpha = 0$, which in turn is connected to singularity of the information matrix at $\alpha = 0$. These problematic features were the motivation to introduce an alternative parameterization which overcomes most if not all of these problems.

It was a pleasant surprise to find that in the present setting the behaviour of the log-likelihood function was to be much more regular, at least for those numerical cases which we have explored. A graphical illustration of this statement is given by Fig. 5 later, which shows some profile log-likelihood plots. These plots refer to a specific data set, but a similar regularity was found with some other data sets which we have considered.

It would be useful to have some theoretical insight into why the log-likelihood function using the skew t-distribution behaves so differently from the SN model, as well as to gather more numerical evidence of its behaviour. However, this theme appears to be a project on its own and cannot be pursued here.

On another front, Fernández and Steel (1999) have highlighted difficulties in regression models when the error term is assumed to have a t-distribution with unspecified degrees of freedom to be estimated from the data. Specifically, their theorem 5 states that there are points of the parameter space where the likelihood function becomes unbounded, if the degrees of freedom are allowed to span over the whole range $\nu \in (0, \infty)$. To avoid this effect, we must restrict the range of ν to the interval (ν_0, ∞), where the threshold ν_0 is a function of X and y. For instance, for a simple random sample with no ties in the y_is, we obtain $\nu_0 = d/(n-1)$, which imposes a very mild limitation. For the stack loss data example discussed by Fernández and Steel (1999) with $d = 1$ and $p = 3$, the value of ν_0 is small, 8/13. In addition, they recalled some numerical examples from the literature where poles have been found by various researchers; in all these cases, however, these poles were found at values of ν that were very small, always below 0.30.

Therefore, in practice the difficulties can be circumvented by avoiding a certain portion of the parameter space which would be somewhat peculiar anyway. However, the fact that ν_0 depends on the response variable leads to a procedure which lacks complete support by the theory of likelihood inference. As advocated by Fernández and Steel (1999), a better theoretical understanding of this sort of model and the associated log-likelihood properties is therefore called for.

It is plausible that regression models with skew t error terms behave quite similarly to analogous cases which employ a regular t-distribution, as for the phenomenon discussed by Fernández and Steel (1999). In the numerical work of the next subsection, we have been driven by considerations described above and decided to ignore poles of the log-likelihood very near $\nu = 0$. We have, however, searched for them, but the only case where we have successfully located one was with the stack loss data, near $\nu = 0.06$, whereas the maximum above the threshold $\nu_0 = 8/13$ was at $\hat{\nu} = 1.14$.

5.2. Numerical examples

5.2.1. Australian Institute of Sport data

It is instructive to examine the outcome of a data fitting process based on the skew t-distribution in a few practical cases. Data on several biomedical variables from 202 athletes have been collected at the Australian Institute of Sport; see Cook and Weisberg (1994) for their description.

We consider here four variables, (BMI, Bfat, ssf, LBM), which represent the body mass index, the percentage of body fat, the sum of skin folds and the lean body mass respectively. An St_4-distribution has been fitted to the 202 points, and Fig. 2 shows the associated Healy plot, using the multivariate normal and the skew t-distribution, as described at the end of Section 4.2. The plots indicate a satisfactory fit to the data provided by the skew t-distribution, which is markedly superior to the normal distribution fit.

Fig. 2 matches Fig. 6 of Azzalini and Capitanio (1999), who fitted an SN distribution to the same data. Although the SN fit was definitely superior to the normal distribution fit, still there was some discrepancy from the identity line which has now vanished almost perfectly.

The full list of estimated parameters is not of particular interest, but it is noteworthy that $\hat{\nu} = 13.7$, which confirms the presence of somewhat longer tails than the normal distribution has.

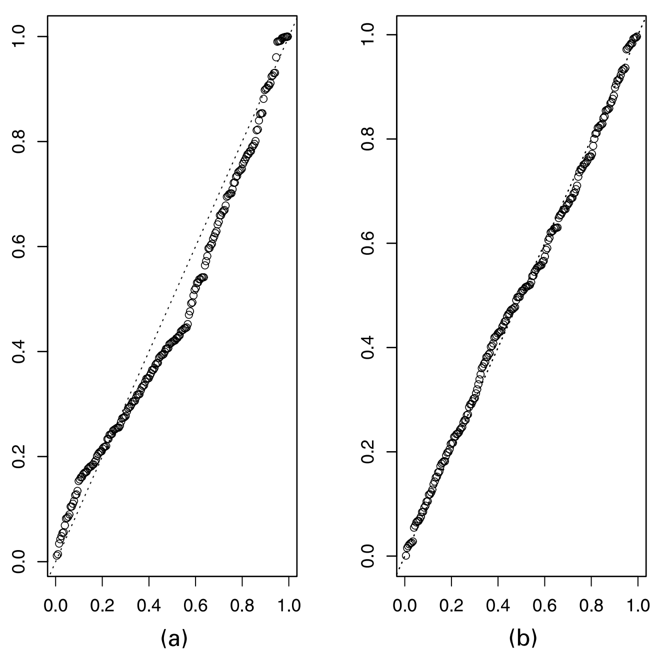

Fig. 2. Australian Institute of Sport data: Healy plot when either (a) a normal distribution or (b) a skew t-distribution is fitted to the data

5.2.2. Martin Marietta data

Our second example considers data taken from Table 1 of Butler *et al.* (1990). On the basis of arguments presented by them, a linear regression is introduced:

$$y = \beta_0 + \beta_1 \text{CRSP} + \varepsilon$$

where y is the excess rate of the Martin Marietta company, CRSP is an index of the excess rate of return for the New York market as a whole and ε is an error term which in our case is taken to be distributed as $\text{St}(0, \omega^2, \alpha)$. Data over a period of $n = 60$ consecutive months are available.

The resulting fitted line is shown in Fig. 3, which displays the scatterplot of the data with superimposed the least squares lines and the line obtained from the above model after adjusting for $\mathbb{E}(\varepsilon)$, whose intercept and slope are

$$\hat{\beta}_0 + \hat{\mathbb{E}}(\varepsilon) = 0.0029,$$
$$\hat{\beta}_1 = 1.248$$

respectively. These values are very close to those obtained by using Jones's (2001) skew t-distribution, and the addition of the corresponding regression line to Fig. 3 would be barely visible, being essentially coincident with our line. The estimated skewness parameter is $\hat{\alpha} \approx 1.246$ with standardized value $1.246/0.653 \approx 1.908$ and observed significance 5.6%. The estimated degrees of freedom are $\hat{\nu} = 3.32$ (standard error 1.43).

As a further indication of the agreement between the observed data and fitted distributions, Fig. 4 shows the histogram of the residuals after removing the line $\hat{\beta}_0 + \hat{\beta}_1 \text{CRSP}$, and the fitted skew t-density; there appears to be a satisfactory agreement between the two.

Other interesting features are indicated by twice the profile log-likelihood functions for the parameters α and $(\alpha, \log(\nu))$ reported in Fig. 5. The contour lines for the two-parameter case are chosen to correspond to differences from the maximum equal to the quantiles of level 0.50, 0.75, 0.90, 0.95 and 0.99 of the χ^2_2-distribution; hence each contoured region can be interpreted

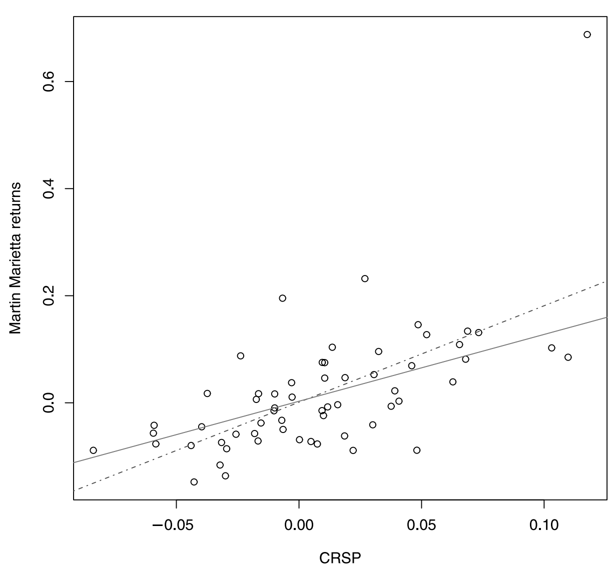

Fig. 3. Martin Marietta data: scatterplot and fitted regression lines; - · - · -, least squares fit; ———, fit using a skew t-error term

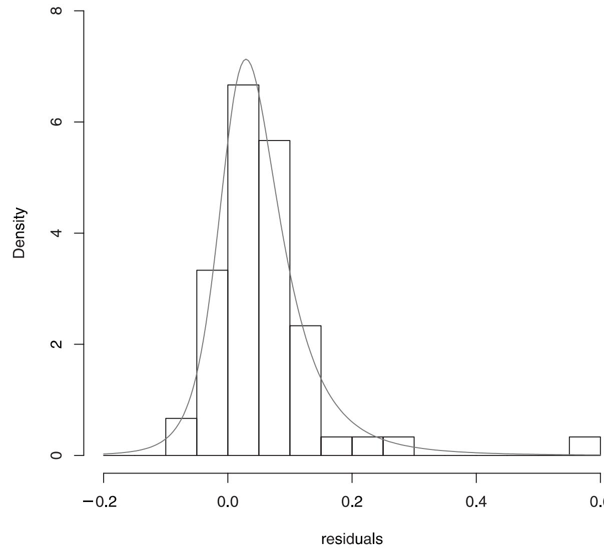

Fig. 4. Martin Marietta data: histogram of the residuals of linear regression and the fitted skew *t*-distribution

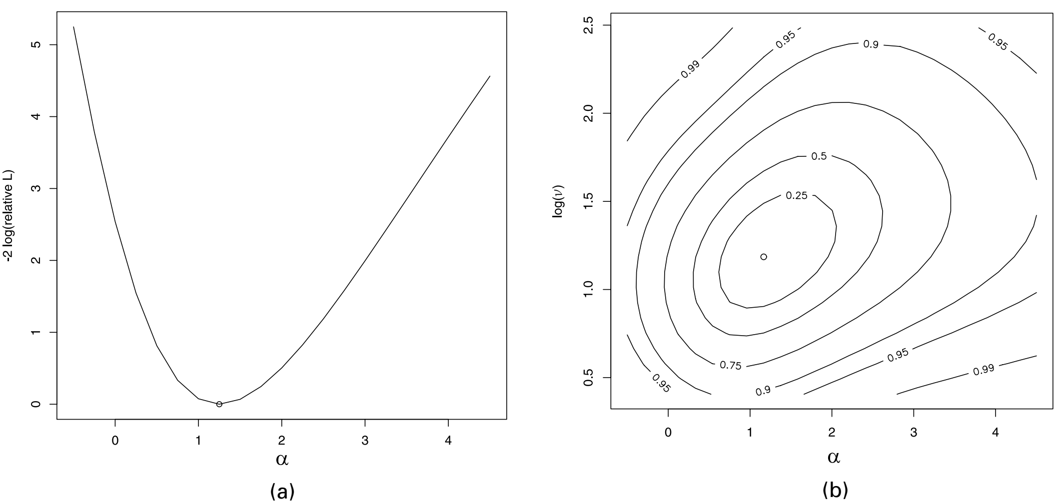

Fig. 5. Martin Marietta data: (a) twice profile negative relative log-likelihood for parameters α and (b) α *versus* $\log(\nu)$

as a confidence region for the pair of parameters, at the confidence level quoted. As anticipated earlier, these plots have quite a regular behaviour, not very far from quadratic functions.

Finally, Fig. 6 compares the Healy plots for the normal and a skew t fitted models. As expected the normal model shows obvious inadequacy, whereas the skew t-model behaves satisfactorily.

6. Discussion

Some broadly related proposals and results have appeared in the recent literature under the connecting concept of the multivariate SN distribution. The present paper has examined the

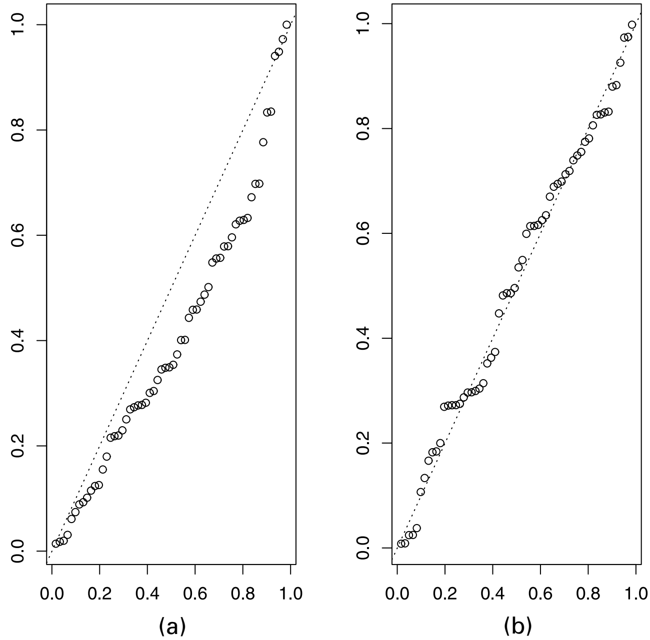

Fig. 6. Martin Marietta data: Healy plot when either (a) a normal distribution or (b) a skew t-distribution is fitted to the data

relationships between many of the above proposals, especially of those dealing with various formulations of the skew elliptical family, by examining their connections and providing a more general approach to obtain several specific results.

Among the broad class of the skew elliptical family, the multivariate skew t-distribution offers ample flexibility for adapting itself to a very wide range of practical situations, and still it maintains mathematical tractability and a set of appealing formal properties. Some numerical evidence and the availability of software developed for inference provide additional support for using the distribution in practical cases. Other interesting distributions have been presented in the literature, most of which fall under the general umbrella of density (12) and its extensions discussed at the end of Section 2.

A wide and closely interconnected set of specific results is evolving towards quite a general framework. There still are open problems, both on the probabilistic and on the inferential side of this area of work, as we have mentioned at various points in the paper, and additional, yet unexpected, results will be discovered. However, what seems to us the more important direction of work, at this stage, is to make use of the available results in tackling real problems. This is the ultimate test to decide about the actual usefulness of all this work.

Acknowledgements

We are grateful to Chris Jones for kindly sending us preliminary versions of his papers, to José Almer Sanqui for drawing our attention to Roberts (1966) and to a referee for helpful comments on earlier versions of this paper. This research has been supported by Ministero dell'Istruzione, dell'Università e della Ricerca, Italy, under grant scheme Progetti di Ricerca di Interesse Nazionale 2000.

References

Arnold, B. C. and Beaver, R. J. (2000a) Hidden truncation models. *Sankhya* A, **62**, 22–35.
Arnold, B. C. and Beaver, R. J. (2000b) The skew Cauchy distribution. *Statist. Probab. Lett.*, **49**, 285–290.
Arnold, B. C. and Beaver, R. J. (2002) Skewed multivariate models related to hidden truncation and/or selective reporting. *Test*, **11**, 7–54.
Azzalini, A. (1985) A class of distribution which includes the normal ones. *Scand. J. Statist.*, **12**, 171–178.
Azzalini, A. and Capitanio, A. (1999) Statistical applications of the multivariate skew normal distribution. *J. R. Statist. Soc.* B, **61**, 579–602.
Azzalini, A. and Dalla Valle, A. (1996) The multivariate skew normal distribution. *Biometrika*, **83**, 715–726.
Branco, M. D. and Dey, D. K. (2001) A general class of multivariate skew elliptical distributions. *J. Multiv. Anal.*, **79**, 99–113.
Butler, R. L., McDonald, J. B., Nelson, R. D and White, S. B. (1990) Robust and partly adaptive estimation of regression models. *Rev. Econ. Statist.*, **72**, 321–327.
Capitanio, A., Azzalini, A. and Stanghellini, E. (2003) Graphical models for skew normal variates. *Scand. J. Statist.*, **30**, in the press.
Cook, R. D. and Weisberg, S. (1994) *An Introduction to Regression Graphics*. New York: Wiley.
David, H. A. (1981) *Order Statistics*, 2nd edn. New York: Wiley.
Fang, K.-T., Kotz, S. and Ng, K. (1990) *Symmetric Multivariate and Related Distributions*. London: Chapman and Hall.
Fernández, C. and Steel, M. F. J. (1998) On Bayesian modelling of fat tails and skewness. *J. Am. Statist. Ass.*, **93**, 359–371.
Fernández, C. and Steel, M. F. J. (1999) Multivariate Student-*t* regression models: pitfalls and inference. *Biometrika*, **86**, 153–168.
Genton, M. G., He, L. and Liu, X. (2001) Moments of skew normal random vectors and their quadratic forms. *Statist. Probab. Lett.*, **51**, 319–325.
Genton, M. G. and Loperfido, N. (2002) Generalized skew-elliptical distributions and their quadratic forms. *Mimeo 2539*. Institute of Statistics, North Carolina State University, Raleigh. (Available from `http://www.stat.ncsu.edu/library/mimeo.html`.)
Genz, A. and Bretz, F. (1999) Numerical computation of multivariate *t*-probabilities with application to power calculation of multiple contrasts. *J. Statist. Computn Simuln*, **63**, 361–378.
Gupta, A. K., Gonzáles-Farías, G. and Domínguez-Molina, J. A. (2001) A multivariate skew normal distribution. *Report I-01-19*. (Available from `http://www.cimat.mx/reportes`.)
Healy, M. J. R. (1968) Multivariate normal plotting. *Appl. Statist.*, **17**, 157–161.
Jones, M. C. (2001) A skew *t* distribution. In *Probability and Statistical Models with Applications: a Volume in Honor of Theophilos Cacoullos* (eds C. A. Charalambides, M. V. Koutras and N. Balakrishnan), pp. 269–278. London: Chapman and Hall.
Jones, M. C. (2002) Multivariate *t* and Beta distributions associated with the multivariate *F* distributions. *Metrika*, **54**, 215–231.
Jones, M. C. and Faddy, M. J. (2003) A skew extension of the *t*-distribution, with applications. *J. R. Statist. Soc.* B, **65**, 159–174.
Kano, Y. (1994) Consistency property of the elliptic probability density functions. *J. Multiv. Anal.*, **51**, 139–147.
Loperfido, N. (2001) Quadratic forms of skew normal random vectors. *Statist. Probab. Lett.*, **54**, 381–387.
Loperfido, N. (2002) Statistical implications of selectively reported inferential results. *Statist. Probab. Lett.*, **56**, 13–22.
Roberts, C. (1966) A correlation model useful in the study of twins. *J. Am. Statist. Ass.*, **61**, 1184–1190.
Sahu, S. K., Dey, D. K. and Branco, M. (2001) A new class of multivariate skew distributions with applications to Bayesian regression models. *Technical Report*. (Available from `http://www.maths.soton.ac.uk/staff/Sahu/research/papers/skew.html`.)
Zuo, Y. and Serfling, R. (2000) On the performance of some robust nonparametric location measures relative to a general notion of multivariate symmetry. *J. Statist. Planng Inf.*, **84**, 55–79.